Plane mirrors

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use the reflection law and locate a virtual image.

SUCCESS CRITERIA - I can...

1. Measure angles from the normal.
2. Use equal incidence and reflection angles.
3. Place the image the same perpendicular distance behind the mirror.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “normal line” and “reflection” to explain your answer. Sentence starters are optional.

normal line: a reference line perpendicular to a surface

reflection: a wave returning from a boundary

virtual image: an image at an apparent origin of rays, not their meeting point

01 Warm-up: bring back an earlier idea

Sensors **1.02 m** apart record delay **0.003 s**. **Find wave speed**. If **0.0005 s** is added to the delay, find the incorrect speed and **describe the bias**.

Sentence starter: The step I would change is … . I would change it because I found … .

Marzano check - circle: ① ② ③

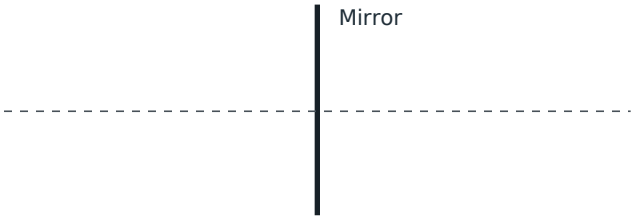
02 Worked example: explain the prediction

Reference: $\theta_{incident} = \theta_{reflected}$
image distance behind = object distance in front.
Plane mirror; **measure angles** from the normal. The image is virtual.

Worked reasoning: An object **2 m** in front of an ideal plane mirror has a virtual image **2 m** behind it. The object-image separation is **4 m**; reflected rays appear to come from the image.

Your turn: Draw the mirror, object, image and backward extensions of two reflected rays.

Sentence starter: In my work, I used … to show … . I labeled it … .



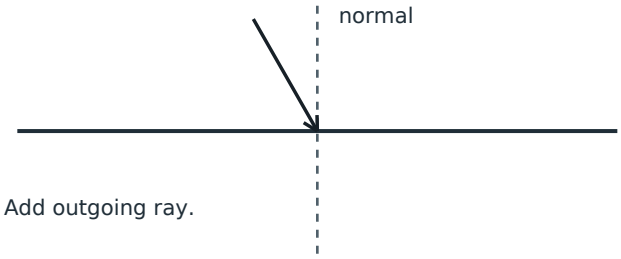
Add object / image or incident / reflected rays.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Draw the normal at the mirror. **Complete a reflected ray** for an incident ray **30°** from normal.

Sentence starter: In my work, I used … to show … . I labeled it … .



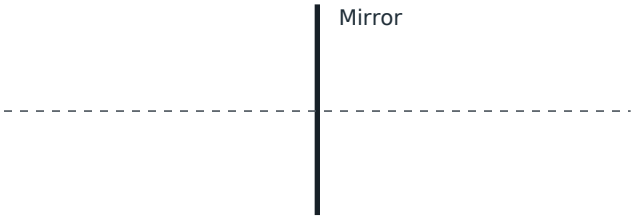
Add outgoing ray.

Marzano check - circle: ① ② ③

04 Apply the physics

A laser ray makes a **20°** angle with a plane mirror’s surface.
1. **Draw the normal** and both rays.
2. **Find incidence angle measured** from the normal, then **the angle** between the incident and reflected ray segments drawn away from the contact point.

Sentence starter: My design includes … . It does what is needed by … .



Add object / image or incident / reflected rays.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

Why is the mirror image called virtual rather than a real object behind the glass?

Sentence starter: For this answer, I treated ... as true. I found ... . This supports ... because ... .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Echoes return at **0.004 s** and **0.006 s** with the same wave speed. **Compare ranges**. Does delay alone find mass?
Label the outgoing and returning paths.

Word help: wave = a disturbance that can carry energy and information

Sentence starter: Both ... . They differ in ... because ... .

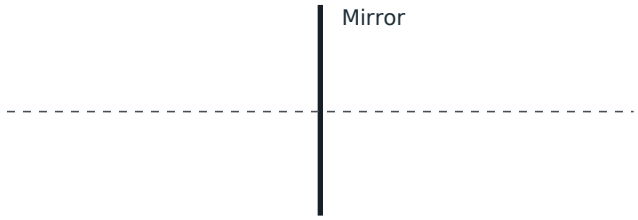


Marzano check - circle: 1 2 3

07 Exit check

An object moves **0.50 m** toward a fixed mirror. **Find image movement** and change in object-image separation.

Sentence starter: I found ... [unit]. This value means ... in this situation.



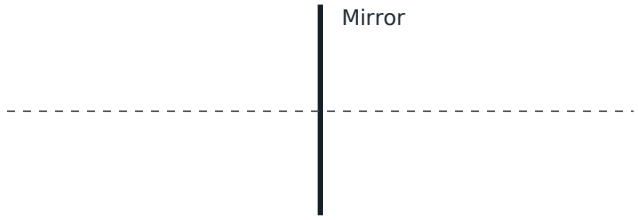
Add object / image or incident / reflected rays.

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

An object moves toward a fixed plane mirror at **1.2 m/s**. **Find image velocity relative** to the mirror and the rate at which object-image separation decreases.

Sentence starter: My design includes ... . It does what is needed by ... .



Add object / image or incident / reflected rays.

Marzano check - circle: 1 2 3